

English	(1001CMD303361240006)		Test Pattern
		CLASSROOM CONTACT PROGRAMME (Academic Session : 2024-2025)	NEET (UG) MINOR
			02-03-2024

PRE-MEDICAL : ENTHUSIAST ADVANCE COURSE PHASE : MED-I (B)

IMPORTANT NOTE : Students having 8 digits **Form No.** must fill two zero before their **Form No.** in OMR. For example, if your **Form No.** is 12345678, then you have to fill **0012345678**.

Test Booklet Code

This Booklet contains **28** pages.

E2

Do not open this Test Booklet until you are asked to do so.

Important Instructions :

- The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on ORIGINAL Copy carefully with **blue/black** ball point pen only.
- The test is of **3 hours 20 minutes** duration and the Test Booklet contains **200** multiple-choice questions (four options with a single correct answer) from **Physics, Chemistry and Biology (Biology-I and Biology-II)**. **50** questions in each subject are divided into **two Sections (A and B)** as per details given below :
 - Section A** shall consist of **35 (Thirty-five)** Questions in each subject (Question Nos - 1 to 35, 51 to 85, 101 to 135 and 151 to 185). All questions are compulsory.
 - Section B** shall consist of **15 (Fifteen)** questions in each subject (Question Nos - 36 to 50, 86 to 100, 136 to 150 and 186 to 200). In Section B, a candidate needs to **attempt any 10 (Ten)** questions out of **15 (Fifteen)** in each subject.

Candidates are advised to read all 15 questions in each subject of Section B before they start attempting the question paper. In the event of a candidate attempting more than ten questions, **the first ten questions answered by the candidate shall be evaluated.**
- Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one mark** will be deducted from the total scores. **The maximum marks are 720.**
- Use **Blue/Black Ball Point Pen only** for writing particulars on this page/markings responses on Answer Sheet.
- Rough work is to be done in the space provided for this purpose in the Test Booklet only.
- On completion of the test, the candidate **must hand over the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator** before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
- The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Form No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
- Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
- Each candidate must show on-demand his/her Allen ID Card to the Invigilator.
- No candidate, without special permission of the Invigilator, would leave his/her seat.
- The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet **twice**. **Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case.**
- Use of Electronic/Manual Calculator is prohibited.
- The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination.
- No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.**
- The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.
- Compensatory time of one hour five minutes will be provided for the examination of three hours and 20 minutes duration, whether such candidate (having a physical limitation to write) uses the facility of scribe or not.

Name of the Candidate (in Capitals) : _____

Form Number : in figures _____

: in words _____

Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ Invigilator's Signature : _____

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SECTION-A (PHYSICS)

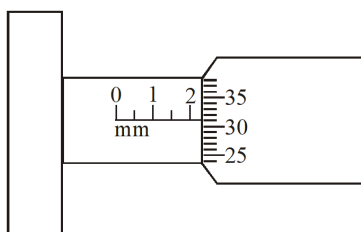
1. Write the dimensions of a/b in the relation

$$P = \frac{a - t^2}{bx}, \text{ where}$$

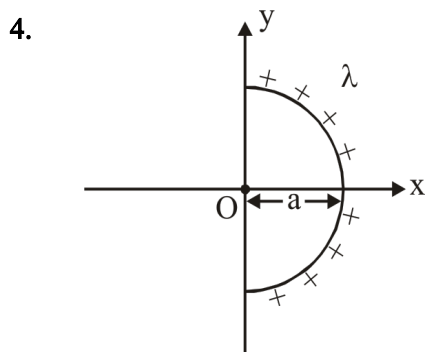
P = pressure, t = time

and x = displacement

- (1) $M^{-1}L^0T^{-2}$ (2) ML^0T^{-2}
 (3) ML^0T^2 (4) MLT^{-2}
2. Round off the number : 3.8060g to three significant digits :-
- (1) 3.81 (2) 3.80
 (3) 3.806 (4) 3.807
3. What is the reading of micrometer screwgauge shown in figure (L.C = 0.01mm) :-



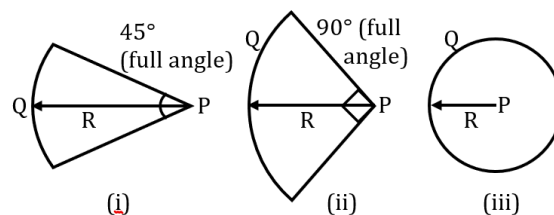
- (1) 2.31 mm (2) 2.29 mm
 (3) 2.36 mm (4) 2.41 mm



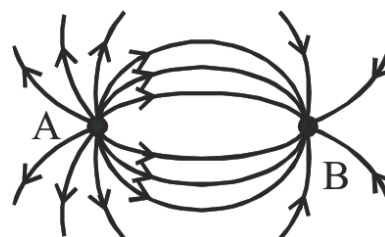
Semi circular ring of radius a is uniformly charged with linear charge density λ . Find out electric field intensity at point O.

- (1) $\frac{\lambda}{2\pi\epsilon_0 a}(-\hat{i})$ (2) $\frac{\lambda}{4\pi\epsilon_0 a}\hat{i}$
 (3) $\frac{\lambda}{2\pi\epsilon_0 a}(+\hat{i})$ (4) $\frac{\lambda}{4\pi\epsilon_0 a}(-\hat{i})$

5. The magnitude of the electric field that is set up at P is minimum in :



- (1) (i)
 (2) (ii)
 (3) (iii)
 (4) equal in all cases
6. The spatial distribution of the electric field due to two charges (A and B) is shown in figure. Which one of the following statements is correct?



- (1) A is +ve and B is -ve and $|A| > |B|$.
 (2) A is -ve and B is +ve and $|A| = |B|$.
 (3) Both are +ve but $A > B$
 (4) Both are -ve but $A > B$
7. Mark the incorrect statement for the electric field lines :
- (1) Field lines start from positive charges and end at negative charges.
 (2) Denser field lines represent weak electric field
 (3) Two field lines can never cross each other.
 (4) Electrostatic field lines never form closed loops.

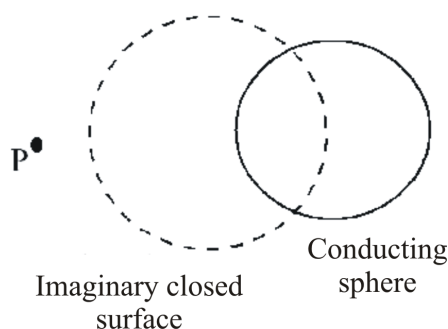
8. A cylinder of radius 'R' and length 'L' is placed in uniform electric field E parallel to the cylinder axis. The total flux for curved surface of the cylinder is given by :

- (1) $E(2\pi R^2)$ (2) $E(\pi R^2)$
(3) $(\pi R^2 L)E$ (4) Zero

9. The electric field in a region of space is given by $E = (5\hat{i} + 2\hat{j})\text{N/C}$. The electric flux due to this field through an area 2m^2 lying in the YZ plane, in S.I. units is:-

- (1) 10 (2) 20
(3) $10\sqrt{2}$ (4) $2\sqrt{29}$

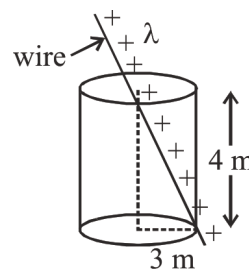
10. Figure shows a imaginary closed surface which intersects a conducting sphere. If a positive charge is placed at the point P, the net flux of the electric field through the imaginary closed surface



- (1) will remain zero
(2) will become positive
(3) will become negative
(4) will become undefined
11. A charge q is placed at one corner of a cube. The electric flux through any of the three faces adjacent to the charge is zero. The flux through any one of the other three faces is :-

- (1) $q/3\epsilon_0$
(2) $q/6\epsilon_0$
(3) $q/12\epsilon_0$
(4) $q/24\epsilon_0$

12. Linear charge density of wire is $8.85\text{ }\mu\text{C/m}$. Radius and height of the cylinder are 3m and 4m. Then find the flux passing through the cylinder :-

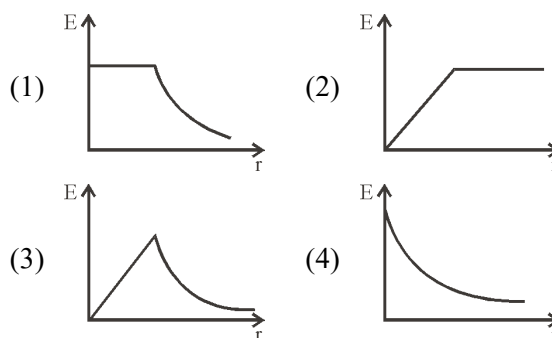


- (1) $5 \times 10^6\text{ volt-m}$ (2) $3 \times 10^6\text{ volt-m}$
(3) $4 \times 10^6\text{ volt-m}$ (4) None

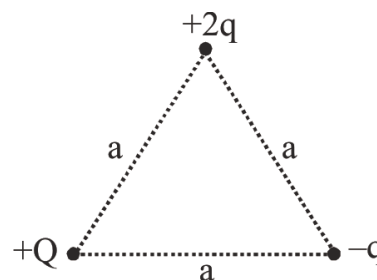
13. The electric field at a distance $\frac{3R}{2}$ from the center of a charged conducting spherical shell of radius R is E. The electric field at a distance R/2 from the center of the sphere is :-

- (1) $\frac{9E}{8}$ (2) Zero (3) $\frac{8E}{4}$ (4) $\frac{E}{3}$

14. Which graph shows variation of electric field of a uniformly charged sphere w.r.t. distance (r) from the centre :-

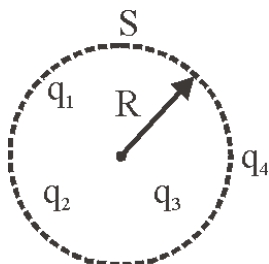


15. Three point charges are placed as shown in figure. If electric potential energy of the system is zero, then find value of Q is.



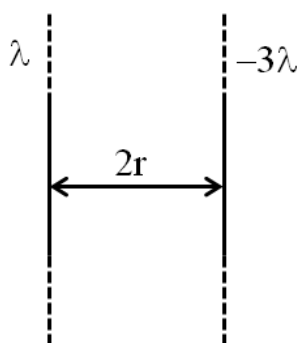
- (1) $+2q$ (2) $-2q$
(3) $+q$ (4) $-q$

16. q_1, q_2, q_3 and q_4 are point charges located at points as shown in the figure and S is a spherical Gaussian surface of radius R . If $\vec{E}_1, \vec{E}_2, \vec{E}_3$ and $\vec{E}_4$ are electric fields due to q_1, q_2, q_3 & q_4 respectively, then which of the following is true according to the Gauss's law :-



- (1) $\oint_S (\vec{E}_1 + \vec{E}_2 + \vec{E}_3) \cdot d\vec{A} = \frac{(q_1 + q_2 + q_3)}{\epsilon_0}$
- (2) $\oint_S (\vec{E}_1 + \vec{E}_2 + \vec{E}_3 + \vec{E}_4) \cdot d\vec{A} = \frac{(q_1 + q_2 + q_3)}{\epsilon_0}$
- (3) $\oint_S (\vec{E}_1 + \vec{E}_2 + \vec{E}_3) \cdot d\vec{A} = \frac{(q_1 + q_2 + q_3 + q_4)}{\epsilon_0}$
- (4) $\oint_S (\vec{E}_1 + \vec{E}_2 + \vec{E}_3) \cdot d\vec{A} = \frac{(q_1 + q_3 + q_4)}{\epsilon_0}$

17. There are two parallel wire having charge density λ and -3λ at $2r$ distance apart, then find electric field at mid point between them.



- (1) $\frac{2K\lambda}{r}$
- (2) $\frac{8K\lambda}{r}$
- (3) $\frac{4K\lambda}{R}$
- (4) $\frac{K\lambda}{r}$

18. A particle 'A' of mass m and charge Q moves directly towards a fixed particle 'B' Which has charge Q . The speed of A is v when it is far away from B. The minimum separation between particles is proportional to :-

- (1) v (2) v^2 (3) v^{-1} (4) v^{-2}

19. $3.2 \times 10^{-7} \text{C}$ charge is distributed uniformly on a conducting sphere having radius 10 cm. Electric field intensity at 30 cm distance from center of sphere will be.

- (1) $3.2 \times 10^4 \text{ N/C}$
- (2) $6.4 \times 10^4 \text{ N/C}$
- (3) $1.6 \times 10^4 \text{ N/C}$
- (4) $12.8 \times 10^4 \text{ N/C}$

20. How does the angle of minimum deviation of a glass prism vary, if the incident violet light is replaced with red light?

- (1) Increases (2) No change
- (3) Decreases (4) None of these

21. If a prism having refractive index $\sqrt{3}$ has angle of minimum deviation equal to the refracting angle of the prism, then the refracting angle of the prism is :-

- (1) 30° (2) 45° (3) 60° (4) 90°

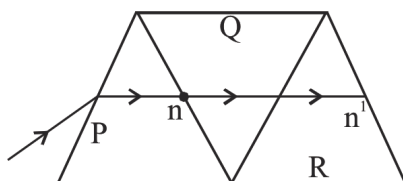
22. The radius of curvature of the convex surface of a thin plano-convex lens is 15 cm and the refractive index of its material is 1.6. The power of the lens is :-

- (1) +1D (2) -2D
- (3) +3D (4) +4D

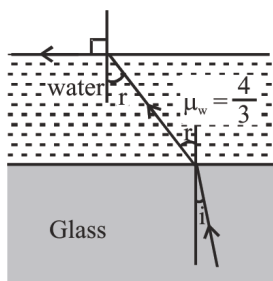
23. Refractive index of prism $\mu = \csc A/2$ then find $\delta_{\min}$:-

- (1) $180-A$ (2) $180-2A$
- (3) $180 - A/2$ (4) $90 - 2A$

24. A given ray of light suffers minimum deviation in an equilateral prism P. Additional prism Q and R of identical shape and of the same material as P are now added as shown in figure. The ray will now suffer :-

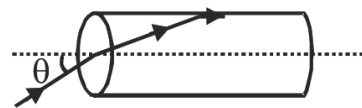


- (1) Greater deviation
 - (2) No deviation
 - (3) Same deviation as before
 - (4) Total internal reflection
25. A convex lens made of glass has focal length 0.15 m in air. If the refractive index of glass is $3/2$ and that of water is $4/3$, the focal length of lens when immersed in water is:
- (1) 0.45 m
 - (2) 0.15 m
 - (3) 0.30 m
 - (4) 0.6 m
26. A ray of light is incident at the glass-water interface at an angle i , it emerges finally parallel to the surface of water, then the value of μ_g would be :-



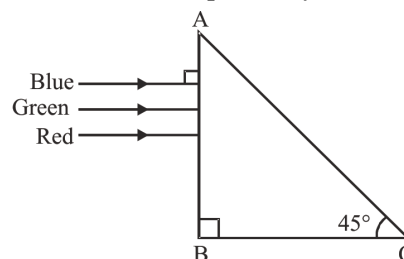
- (1) $(4/3)/\sin i$
- (2) $1/\sin i$
- (3) $4/3$
- (4) 1

27. A transparent solid cylindrical rod has a refractive index of $2/\sqrt{3}$. It is surrounded by air. A light ray is incident at the mid-point of one end of the rod as shown in the figure.



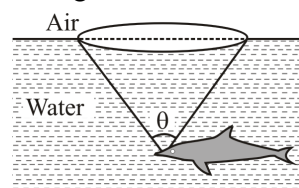
The incident angle θ for which the light ray grazes along the wall of the rod is :-

- (1) $\sin^{-1}(1/2)$
 - (2) $\sin^{-1}(\sqrt{3}/2)$
 - (3) $\sin^{-1}(2/\sqrt{3})$
 - (4) $\sin^{-1}(1/\sqrt{3})$
28. A beam of light consisting of red, green and blue colours is incident on a right angled prism. The refractive index of the material of the prism for the above red, green and blue wavelengths are 1.39, 1.44 and 1.47, respectively.



The prism will :-

- (1) separate the red colour part from the green and blue colours
 - (2) separate the blue colour part from the red and green colours
 - (3) separate all the three colours from one another
 - (4) not separate the three colours at all
29. A fish is a little away below the surface of a lake. If the critical angle is 49° , then the fish could see things above the water surface within a plane angular range of θ° . The value of ' θ ' is



- (1) $\theta = 49^\circ$
- (2) $\theta = 90^\circ$
- (3) $\theta = 98^\circ$
- (4) $\theta = 24\frac{1}{2}^\circ$

30. A convex lens of focal length 40 cm and a concave lens of focal length 20 cm is combined. The power of combination is :-

- (1) 5 D (2) 2 D
(3) -2.5 D (4) 7.5 D

31. An object is placed at a distance of $\frac{f}{2}$ from a convex lens of focal length f . The image will be

- (1) At one of the foci, virtual and double its size
(2) Is greater than 1.5 but less than 2.0
(3) At $2f$, virtual and erect
(4) None of the above

32. A convex lens A of focal length 20 cm and a concave lens B of focal length 5 cm are kept along the same axis with a distance d between them. If a parallel beam of light falling on A leaves B as a parallel beam, then the distance d (in cm) will be

- (1) 20 (2) 15
(3) 30 (4) 50

33. A beam of converging light converges towards a point. In the path of light rays, a convex lens of focal length 20 cm is placed at a distance 10 cm from converging point. Find position where light rays coverage after passing through lens.

- (1) 5 cm
(2) 10 cm
(3) 6.67 cm
(4) 8 cm

34. If two lenses of +5 diopters are mounted at some distance apart, the equivalent power will always be negative if the distance is :-

- (1) Greater than 40 cm
(2) Equal to 40 cm
(3) Equal to 10 cm
(4) Less than 10 cm

35. A convex lens is dipped in a liquid whose refractive index is equal to the refractive index of the lens. Then its focal length will -

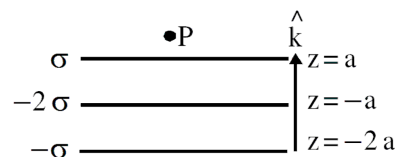
- (1) Become zero
(2) Become infinite
(3) Become small, but non-zero
(4) Remain unchanged

SECTION-B (PHYSICS)

36. If force (F), work (W) and velocity (v) are taken as fundamental quantities, then the dimensional formula of time (T) is :-

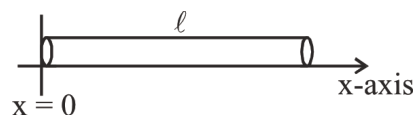
- (1) $[WFv]$ (2) $[WFv^{-1}]$
(3) $[W^{-1}F^{-1}v^{-1}]$ (4) $[WF^{-1}v^{-1}]$

37. Three large parallel sheets have uniform surface charge densities as shown in the figure. What is the electric field at P ?



- (1) $-\frac{4\sigma}{\epsilon_0}\hat{k}$ (2) $\frac{4\sigma}{\epsilon_0}\hat{k}$
(3) $-\frac{2\sigma}{\epsilon_0}\hat{k}$ (4) $-\frac{\sigma}{\epsilon_0}\hat{k}$

38. Consider a rod of length ℓ having linear charge density ($\lambda = \lambda_0 x^2$) placed along x-axis whose one end is placed at origin. Find charge on the rod :-

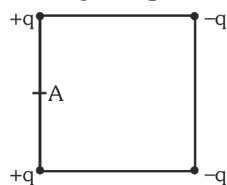


- (1) $\frac{\lambda_0 \ell^3}{3}$
(2) $\frac{\lambda_0 \ell^4}{4}$
(3) $\frac{\lambda_0 \ell^2}{2}$
(4) $\frac{\lambda_0 \ell^3}{4}$

39. Column I specifies a point distance r from the center/axis of a symmetrical distribution of charge. Column II gives the variation of electric field intensity at P as a function of r . Match the columns :-

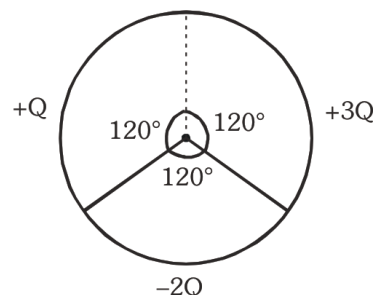
Column I		Column II	
i	P lies outside a straight wire having uniform linear charge density	a	$E \propto \frac{1}{r^2}$
ii	P lies inside a spherical charged conductor	b	$E \propto \frac{1}{r}$
iii	P lies inside a spherical body having uniform volume charge density	c	$E \propto r$
iv	P lies outside the plane non-conducting sheet	d	$E \propto r^0$

- (1) i-b, ii-d, iii-c, iv-d (2) i-a, ii-b, iii-c, iv-d
 (3) i-b, ii-d, iii-a, iv-c (4) i-d, ii-b, iii-a, iv-c
40. At a distance of 1 m from a fixed charge of 4mC , a particle of mass 2 g and charge $1\mu\text{C}$ is held stationary. Both the charges are placed on a smooth horizontal surface. If the particle is made free to move, then its speed at a distance of 10 m from the fixed charge will be :-
- (1) 180 ms^{-1} (2) 20 ms^{-1}
 (3) 60 ms^{-1} (4) 90 ms^{-1}
41. Four electric charges $+q$, $+q$, $-q$ and $-q$ are placed at the corners of a square of side $2L$ (see figure). The electric potential at point A, midway between the two charges $+q$ and $+q$, is :-

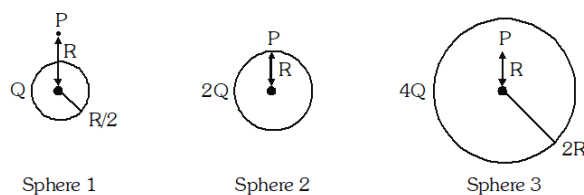


- (1) $\frac{1}{4\pi\epsilon_0} \frac{2q}{L} (1 + \sqrt{5})$
 (2) $\frac{1}{4\pi\epsilon_0} \frac{2q}{L} \left(1 + \frac{1}{\sqrt{5}}\right)$
 (3) $\frac{1}{4\pi\epsilon_0} \frac{2q}{L} \left(1 - \frac{1}{\sqrt{5}}\right)$
 (4) Zero

42. Figure shows three circular arcs, each of radius R and total charge as indicated. The net electric potential at the centre of curvature is :-

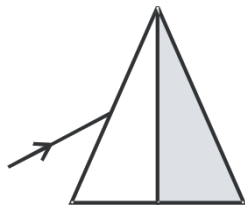


- (1) $\frac{Q}{4\pi\epsilon_0 R}$
 (2) $\frac{Q}{2\pi\epsilon_0 R}$
 (3) $\frac{2Q}{\pi\epsilon_0 R}$
 (4) $\frac{Q}{\pi\epsilon_0 R}$
43. Charges Q , $2Q$ and $4Q$ are uniformly distributed in three dielectric solid spheres 1, 2 and 3 of radii $R/2$, R and $2R$ respectively, as shown in figure. If magnitudes of the electric fields at point P at a distance R from the centre of spheres 1, 2 and 3 are E_1 , E_2 and E_3 respectively, then

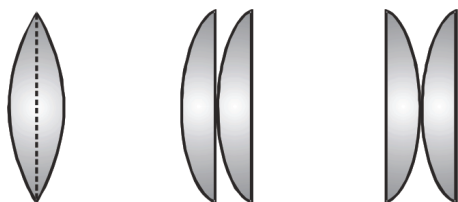


- (1) $E_1 > E_2 > E_3$
 (2) $E_3 > E_1 > E_2$
 (3) $E_2 > E_1 > E_3$
 (4) $E_3 > E_2 > E_1$

44. A light ray is incident upon an equilateral prism in minimum deviation position and suffers a deviation of 34° . If the shaded half of the prism is knocked off, the ray will ?



- (1) Suffer a deviation of 34°
 (2) Suffer a deviation of 68°
 (3) Suffer a deviation of 17°
 (4) Not come out of the prism
45. Two similar plano-convex lenses are combined together in three different ways as shown in the figure. The ratio of the focal lengths in three cases will be :-



- (1) $2 : 2 : 1$
 (2) $1 : 1 : 1$
 (3) $1 : 2 : 2$
 (4) $2 : 1 : 1$
46. The refractive index of a particular material is 1.67 for blue light, 1.65 for yellow light and 1.63 for red light, Dispersive power of the material is :-
- (1) 0.024
 (2) 0.031
 (3) 0.0615
 (4) 0.6015

47. A converging lens having magnitude of focal length as f_1 is kept coaxially in contact with a diverging lens having magnitude of focal length as f_2 . The focal length of the combination would be :

(1) $\frac{f_1 f_2}{f_2 - f_1}$ (2) $\frac{f_1 + f_2}{f_1 f_2}$
 (3) $\frac{f_1 - f_2}{f_1 f_2}$ (4) $\frac{f_1 f_2}{f_1 + f_2}$

48. Two identical glass ($\mu_g = \frac{3}{2}$) equiconvex lens of focal length 20 cm are kept in contact. The space in between the two lenses is filled with water ($\mu_w = \frac{4}{3}$). The focal length of the combination is:-

(1) 60 cm (2) 30 cm
 (3) 20 cm (4) 15 cm

49. A ray of light traveling in water is incident on its surface open to air. The angle of incidence is θ , which is less than the critical angle. Then there will be :

- (1) Only a reflected ray and no refracted ray
 (2) Only a refracted ray and no reflected ray
 (3) A reflected ray and a refracted ray and the angle between them would be less than $180^\circ - 2\theta$
 (4) A reflected ray and a refracted ray and the angle between them would be greater than $180^\circ - 2\theta$.

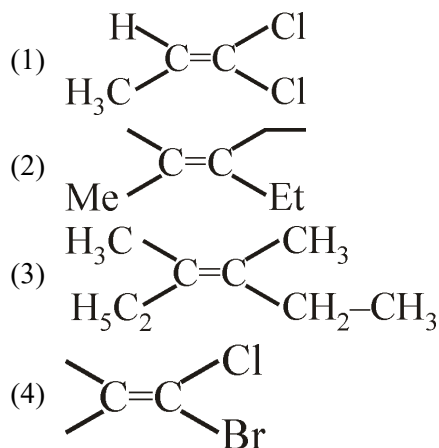
50. A ray of light is incident at angle i on a surface of a prism of small angle A and emerges normally from the opposite surface. If the refractive index of the material of the prism is μ , the angle of incidence i is nearly equal to :

(1) A/μ (2) $A/(2\mu)$
 (3) μA (4) $\mu A/2$

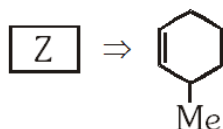
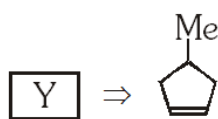
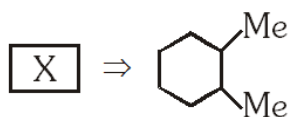
Topic : SOLUTION, ISOMERISM

SECTION-A (CHEMISTRY)

51. Which of the following can show geometrical isomerism ?

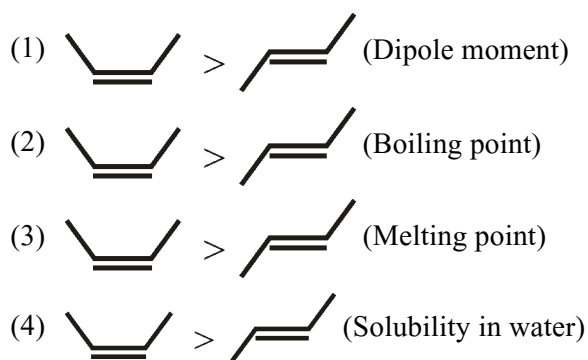


52. Number of chiral carbon atoms in the compound X, Y and Z respectively would be :

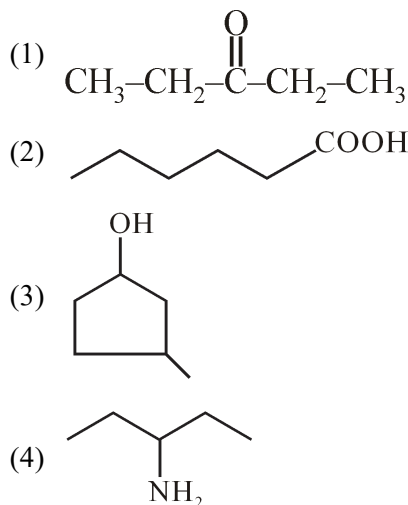


- (1) 2, 1, 1 (2) 1, 1, 1
(3) 2, 0, 2 (4) 2, 0, 1

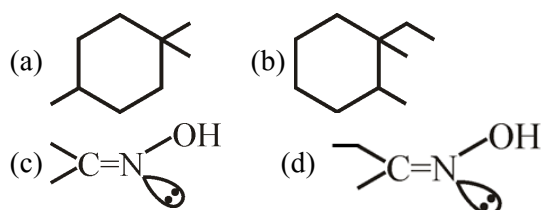
53. Select incorrect match with respect to cis but-2-ene and trans but-2-ene.



54. In which of the following compound metamerism is possible :

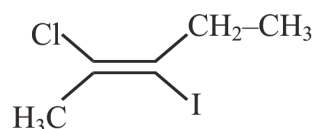


55. Which of the following will show geometrical isomerism ?



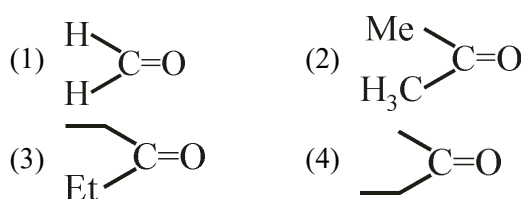
- (1) a, b, d (2) b, c
(3) b, d (4) b, c, d

56. The complete IUPAC name of following compound ?



- (1) Cis-2-chloro-3-iodopent-2-ene
(2) Trans-2-chloro-3-iodopent-2-ene
(3) Cis-3-iodo-4-chloropent-3-ene
(4) Trans-3-iodo-4-chloropent-3-ene

57. Oxime of which of the following compound show geometrical isomerism ?



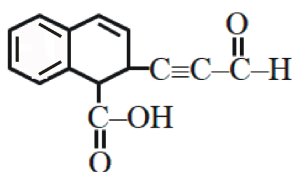
58. 2-butyne and 1, 3-butadiene are :

- (1) Chain isomer (2) Position isomer
(3) Functional isomer (4) Tautomers

59. Isomers have essential identical :-

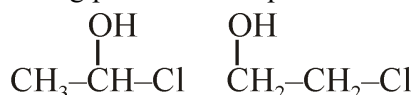
- (1) Structural formula
(2) Functional group
(3) Molecular formula
(4) Principle carbon chain

60. Degree of unsaturated of given compound is-



- (1) 12 (2) 6 (3) 8 (4) 10

61. The following pair related as position isomer



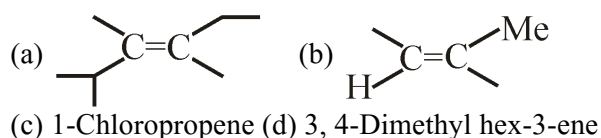
The differ in position of :-

- (1) Principle functional group
(2) Substituent
(3) Both principle functional group and substituent
(4) Methyl group

62. Minimum number of carbon atom should be present in alkane, alkene and alkyne to show positional isomerism is, respectively-

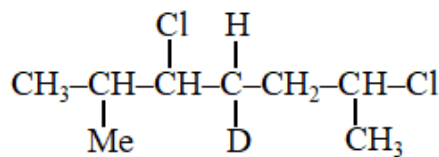
- (1) 4, 4, 5 (2) 6, 5, 5
(3) 6, 4, 4 (4) 5, 4, 4

63. Which of the following compounds show geometrical isomerism?



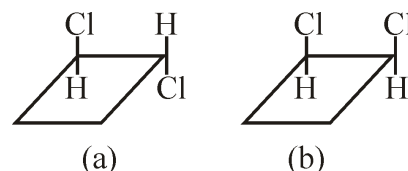
- (1) a, b, c, d (2) a, c, d
(3) a, b, d (4) b, c, d

64. Number of chiral carbon atoms in given compound is-



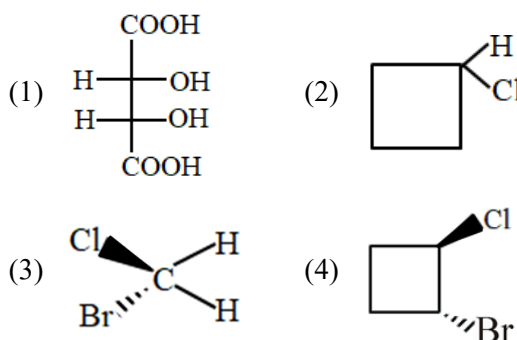
- (1) 4 (2) 3 (3) 2 (4) 1

65. Select correct statement-



- (1) (a) has trans configuration and b has cis configuration
(2) (a) and (b) are geometrical isomers
(3) (a) and (b) can show geometrical isomerism
(4) All are correct

66. Which of the following is optically active-



67. Total number of isomeric alkene possible from C_4H_8 (Structural isomers only)

- (1) 5 (2) 2 (3) 3 (4) 4

68. A solution is prepared by adding 2g of a substance A (solute) to 18g water. What will be the % w/w of solute in the solution :-

- (1) 10 % (2) 40 %
(3) 11.11 % (4) None of these

69. The number of mole of NaOH present 0.2 molal aqueous solution containing 150 g water is :-

- (1) 0.2 (2) 0.03 (3) 0.003 (4) 0.01

70. The volume strength of 1.5 N H_2O_2 solution is :

- (1) 4.8 (2) 8.4 (3) 3.0 (4) 8.0

71. The value of vapour pressure of solution which contain benzene and octane in molar ratio of 4 : 3 is-

(V.P. of Benzene = 300 mm Hg)

(V.P. of Octane = 400 mm Hg)

- (1) 310 mmHg (2) 358 mmHg

- (3) 325 mmHg (4) 343 mmHg

72. If P_A° and P_B° are 108 and 36 torr respectively. What will be the mole fraction of A in vapour phase if B has mole fraction in solution 0.5.

- (1) 0.25 (2) 0.75

- (3) 0.60 (4) 0.35

73. **Assertion** : Molarity of a solution in liquid state changes with temperature.

Reason : The volume of a solution changes with change in temperature.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

74. **Assertion** : One molal aqueous solution of glucose contains 180 g of glucose in 1kg water.

Reason : Solution containing one mole of solute in 1000 g of solvent is called one molal solution.

(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(2) (A) is correct but (R) is not correct

(3) (A) is incorrect but (R) is correct

(4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

75. **Assertion (A)** : When non-volatile solute is added to solvent, the vapour pressure of the solution decreases.

Reason (R) : As number of solvent molecules escaping from the surface is reduced, the vapour pressure of the solution is also reduced.

(1) Both A and R are correct and R is the correct explanation of A.

(2) Both A and R are correct but R is not the correct explanation of A.

(3) A is correct but R is incorrect.

(4) A is incorrect but R is correct.

76. The observed & calculated molar mass of $\text{Ba}(\text{NO}_3)_2$ is 91 and 261 respectively. Calculate its degree of ionisation?

- (1) 80% (2) 93%

- (3) 70% (4) 87.5%

77. Match the column

Column - I (Solute)		Column-II (Van't Hoff factor, i)	
(A)	AlCl_3 if $\alpha = 0.8$	(P)	$i = 3.4$
(B)	BaCl_2 if $\alpha = 0.9$	(Q)	$i = 2.8$
(C)	CH_3COOH dissolved in benzene	(R)	$i = 3.8$
(D)	$\text{K}_4[\text{Fe}(\text{CN})_6]$, if $\alpha = 0.7$	(S)	$i = 0.5$

(1) $\text{A} \rightarrow \text{P}$, $\text{B} \rightarrow \text{Q}$, $\text{C} \rightarrow \text{R}$, $\text{D} \rightarrow \text{S}$

(2) $\text{A} \rightarrow \text{Q}$, $\text{B} \rightarrow \text{P}$, $\text{C} \rightarrow \text{S}$, $\text{D} \rightarrow \text{R}$

(3) $\text{A} \rightarrow \text{P}$, $\text{B} \rightarrow \text{Q}$, $\text{C} \rightarrow \text{S}$, $\text{D} \rightarrow \text{R}$

(4) $\text{A} \rightarrow \text{R}$, $\text{B} \rightarrow \text{S}$, $\text{C} \rightarrow \text{P}$, $\text{D} \rightarrow \text{Q}$

78. 80 g NaOH is present in 500 mL aqueous solution having density 1.8 g/mL then the molality of solution is :-

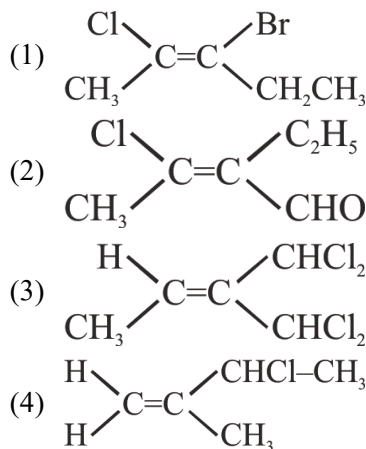
- (1) 2.43 m (2) 4 m

- (3) 1.31 m (4) 1.8 m

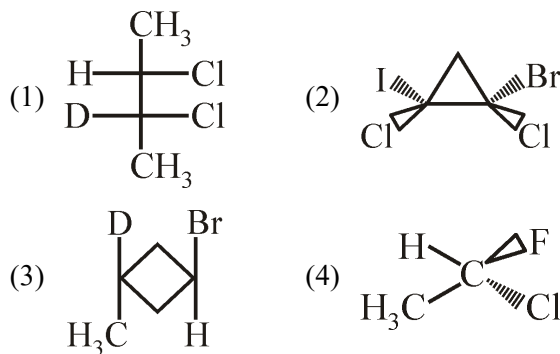
SECTION-B (CHEMISTRY)

79. Vapour pressure of aqueous solution of urea is 700 mm Hg at 100°C then what is the mole fraction of urea?
- (1) 0.078 (2) 0.922
(3) 0.038 (4) 0.058
80. 1000 g aqueous solution of CaCO_3 contains 18 g of CaCO_3 then the concentration of solution is :
- (1) 18 ppm (2) 18×10^3 ppm
(3) 18×10^{-3} ppm (4) 9×10^3 ppm
81. What is the mole fraction of solute in 2 molal aqueous solution?
- (1) 0.34 (2) 0.034
(3) 0.017 (4) 0.05
82. What is the molality of solution in which HCl is solute having mole fraction 0.7 and water is solvent?
- (1) 63.92 m (2) 23.80 m
(3) 11.74 m (4) 129.6 m
83. How much water should be added to 500 cc of centinormal solution of H_2SO_4 to make it exactly millinormal?
- (1) 4500 cc (2) 5000 cc
(3) 4000 cc (4) 3500 cc
84. What is the molarity of 0.02 N potash Alum solution ?
- (1) 2.5×10^{-4} M
(2) 1.6×10^{-6} M
(3) 0.01 M
(4) 2.5×10^{-3} M
85. The normality of 60% CH_3COOH solution [$d = 1.5 \text{ g/mL}$] by wt. is :-
- (1) 30 N (2) 90 N
(3) 15 N (4) 7.5 N

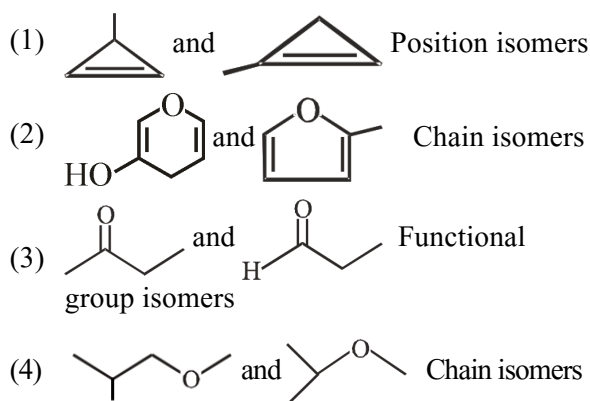
86. Select the E-isomer among the following :-



87. In which of the following compound plane of symmetry is exist ?



88. Which is correctly matched ?



89. and are :-

- (1) Position isomer
(2) Functional group isomer
(3) Chain isomer
(4) None

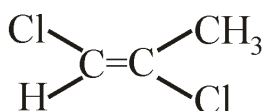
90. How many structural isomers of ether can be formed from molecular formula $C_5H_{12}O$.

- (1) 2 (2) 4 (3) 6 (4) 14

91. $\begin{array}{c} \text{CH}_2-\text{CH}_2-\text{CHO} \\ | \\ \text{CN} \end{array}$ and $\begin{array}{c} \text{CH}_2-\text{CH}_2-\text{CHO} \\ | \\ \text{NC} \end{array}$

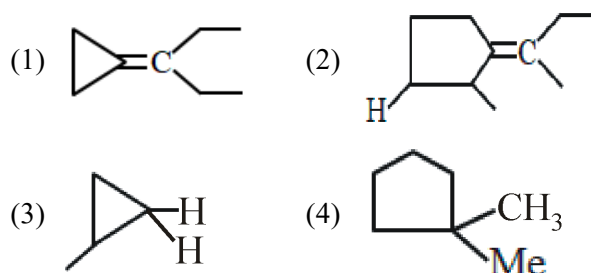
- (1) Chain isomers
(2) Identical
(3) Functional group isomers
(4) Not isomers

92. Select correct configuration-



- (1) E, Trans (2) E, Cis
(3) Z, Cis (4) Z, Trans

93. Which of the following can show geometrical isomerism.



94. The Van't Hoff factor for 0.1M $Ba(NO_3)_2$ solution is 2.74. The degree of dissociation is :-

- (1) 91.3% (2) 87%
(3) 100% (4) 74%

95. Which of the following concentration term is temperature independent -

- (A) % w/w (B) Molarity
(C) Mole fraction (D) Molality

- (1) Only D
(2) A and B
(3) A, C and D
(4) Only B

96. **Assertion** : Amalgam of mercury with sodium is an example of solid solutions.

Reason : Mercury is solvent and sodium is solute in the solution.

- (1) Both **assertion** and **reason** are true and **reason** is the correct explanation of **assertion**.
(2) Both **assertion** and **reason** are true but **reason** is NOT the correct explanation of **assertion**.
(3) **Assertion** is true but **reason** is false.
(4) Both **assertion** and **reason** are false.

97. A solution of liquids A and B having vapour pressure in pure state P_A^0 and P_B^0 . The solution contains 30% mole of A which is in equilibrium with 60% mole of A in vapour phase. If P_B^0 is 2 cm of Hg, the P_A^0 is :-

- (1) 70 cm of Hg (2) 7 cm of Hg
(3) 700 cm of Hg (4) 65 cm of Hg

98. How much mass of solute [Molar mass = 30] is required to dissolve in 100 g heptane to reduce its vapour pressure to 70% ?

- (1) 14 g (2) 12.85 g
(3) 10 g (4) 20 g

99. Vapour pressure of pure benzene is 800 mm Hg. If 3g of non-volatile solute is added in 19.5 g of benzene, the vapour pressure is reduced to 760 mm Hg. Calculate molar mass of solute?

- (1) 128 g (2) 300 g
(3) 400 g (4) 228 g

100. The molarity of 800 g of pure water is :-

- (1) 20 M
(2) 55.5 M
(3) 4 M
(4) Can't be determined

Topic : MOLECULAR BASIS OF INHERITANCE (COMPLETE), HUMAN REPRODUCTION, REPRODUCTIVE HEALTH, HUMAN HEALTH AND DISEASE

Join Telegram: @NEETxNOAH

SECTION-A (BIOLOGY-I)

101. Amino acid binding site in tRNA is :-

- (1) CCA 3' end (2) DHU loop
(3) 5' end (4) Anticodon loop

102. The transacetylase of the lac operon is synthesized from :

- (1) i gene
(2) z gene
(3) y gene
(4) a gene

103. Transcription occurs in the presence of :

- (1) DNA polymerase-I
(2) DNA polymerase-II
(3) RNA polymerase
(4) DNA polymerase

104. The lac operon concept was proposed by

- (1) Watson and Crick (2) H.G. Khorana
(3) Jacob and Monad (4) Meselson & Stahl

105. Gene is a :

- (1) RNA molecule (2) Protein molecule
(3) DNA molecule (4) Histone protein

106. Which of the following is associated with nucleolus ?

- (1) rRNA (2) mRNA
(3) tRNA (4) DNA

107. In absence of lactose, the repressor protein in case of lac operon will bind to

- (1) Promoter site (2) Operator site
(3) Regulator site (4) Structural gene

108. c-DNA probes are copied from messenger RNA molecules with the help of :

- (1) Restriction enzyme
(2) Reverse transcriptase
(3) DNA dependent DNA polymerase
(4) RNA polymerase

109. Find the incorrect statement for eukaryotic transcription :-

- (1) Introns are spliced and exons are joined
(2) Polyadenylation is done at 3' end
(3) Translation begins before the completion of transcription
(4) Takes place inside nucleus

110. If one strand of DNA has nitrogenous base as 5'-ATCTG-3'. What would be the complementary RNA stand sequence ?

- (1) 5'-AUCUG-3' (2) 5' CAGAU-3'
(3) 3' UAGAC-5' (4) 3' ATCGU 5'

111. The accessibility of promotor regions of prokaryotic DNA by RNA polymerase is in many cases regulated by the interaction of some proteins with sequences termed as :-

- (1) Operator (2) Promotor
(3) Regulator (4) Terminator

112. Insertion or deletion of one or two bases changes the reading frame from the point of insertion or deletion but insertion or deletion of three or its multiple bases insert or delete one or multiple codon. This forms the genetic basis of proof that codon is a and it is read in a manner

- (1) Triplet, contiguous
(2) Unambiguous, contiguous
(3) Triplet, overlapping
(4) Triplet, degenerate

113. Select the group of three codons, which do not code for any amino acid and result in the termination of polypeptide chain synthesis :-

- (1) UAA, UAG, UGG
- (2) UAA, UAG, UGA
- (3) UGA, UGU, UAA
- (4) UAA, UAG, UUU

114. Read the following four statements A to D and choose the correct option :

- (A) RNA was the first genetic material
- (B) RNA used to act as a genetic material as well as a catalyst
- (C) RNA being a catalyst was reactive and hence unstable
- (D) RNA is evolved from DNA

How many statements are correct :

- (1) A, B, C
- (2) A, B
- (3) C, D
- (4) B, C

115. Match column-I to column-II

Codon Amino acid

- (i) AUG (a) Valine
- (ii) GUG (b) Methionine
- (iii) AAA (c) Phenylalanine
- (iv) UUU (d) Lysine

- (1) i-a, ii-b, iii-c, iv-d
- (2) i-d, ii-b, iii-c, iv-a
- (3) i-b, ii-a, iii-d, iv-c
- (4) i-c, ii-a, iii-d, iv-c

116. Which type of ribosomal RNA is not found in cytosol of eukaryotes ?

- (1) 23s (2) 18s (3) 5s (4) 5.8s

117. Which of the following is required by RNA polymerase for transcription initiation ?

- (1) Sigma factor (2) Beta subunit
- (3) Rho factor (4) Spliceosome

118. Portion of gene which is transcribed but not translated is :

- (1) Exon (2) Intron
- (3) Cistron (4) Codon

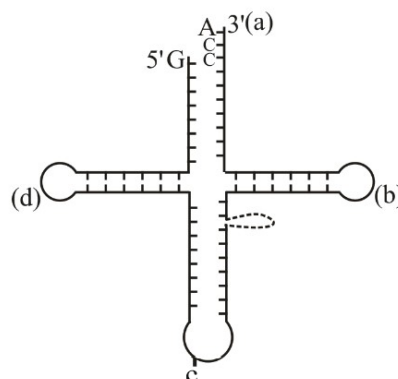
119. Which of the following RNA act as a template for protein synthesis ?

- (1) r-RNA (2) m-RNA
- (3) t-RNA (4) mi-RNA

120. Which is not correct about translation :

- (1) It starts with AUG
- (2) Stopped at termination codon
- (3) Occurs in nucleus
- (4) Dictated by DNA and represented by mRNA

121.



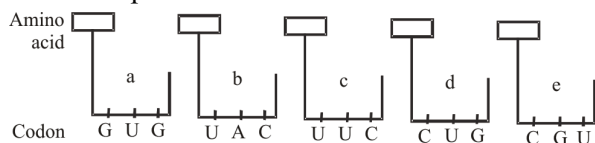
Identify (a), (b), (c) and (d) in the given structure :-

	(a)	(b)	(c)	(d)
(1)	Acceptor end	TΨC loop	NODOC	Attachment loop
(2)	Acceptor end	TΨC loop	Anticodon	Amino-acyl synthetase recognition loop
(3)	Attachment loop	TΨC loop	NODOC	DHU loop
(4)	Recognition loop	DHU loop	Anticodon	Acceptor end

122. Which one of the following can give a complimentary and palindromic sequence :

- (1) 5' ATATCC 3' (2) 5' CCGAAT 3'
(3) 5' GAATTC 3' (4) 5' AGGTTC 3'

123. Find the correct sequence of binding of the following amino-acyl-tRNA complex during translation of a mRNA transcribed by a DNA segment having following base sequence
3' TAC GTG CTG TTC CGT 5' Template strand
Select the correct answer showing the correct order of alphabet



- (1) b, a, d, c, e (2) a, b, c, d, e
(3) d, c, b, a, e (4) d, b, c, a, e

124. Select the correct statements :-

- (A) Chromosomal aberrations are commonly observed in cancer cells.
(B) Sickle cell anaemia is an example of transversion and mis-sense mutation.
(C) Deletions and insertions of base pairs of DNA cause frame shift mutation.
(D) Mutation cause changes in the genotype and phenotype of an organism.

Options :-

- (1) A, B and C only (2) A, B and D only
(3) A, C, and D only (4) A, B, C and D

125. How many of the following statements are correct ?

- (A) In prokaryotes, control of the rate of transcriptional initiation is the predominant site for control of gene expression.
(B) Process of splicing represents the dominance of DNA world
(C) Presence of intron is reminiscent of antiquity
(D) Inheritance of a character is also affected by promoter and regulatory sequences of a structural gene.

- (1) 2 (2) 3 (3) 4 (4) 5

126. Study the lists and find the correct match :

List I		List II	
a.	Exon	I.	Site for binding of RNA polymerase
b.	Capping	II.	Coding sequence
c.	Tailing	III.	Lagging strand
d.	Promoter	IV.	Methyl guanosine triphosphate
		V.	Adenylate residues

- (1) a-II, b-IV, c-V, d-I
(2) a-II, b-IV, c-I, d-V
(3) a-III, b-I, c-II, d-IV
(4) a-IV, b-II, c-III, d-I

127. Lac operon in prokaryotes :-

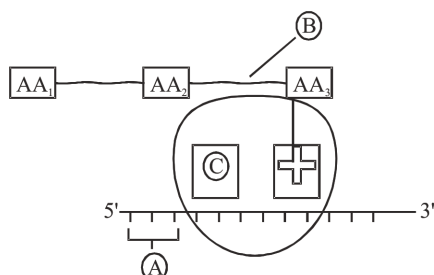
- (a) is regulated negatively
(b) is inducible operon
(c) cannot be regulated positively
(1) Both (a) and (b) are correct
(2) only (c) is correct
(3) Only (b) is incorrect
(4) Both (b) and (c) are incorrect

128. Match column-I with column-II and choose correct answer

	Column-I		Column-II
(a)	RNA polymerase-I	(i)	hn-RNA
(b)	RNA polymerase-II	(ii)	t-RNA
(c)	RNA polymerase-III	(iii)	r-RNA

- (1) a-i, b-ii, c-iii
(2) a-ii, b-i, c-iii
(3) a-iii, b-i, c-ii
(4) a-iii, b-ii, c-i

129.



Identify A, B, C :-

	A	B	C
(1)	GUG	H-bond	T-site
(2)	AUG	Peptide bond	P-site
(3)	AUG	Peptide bond	T-site
(4)	SD sequence	Peptide bond	P-sit

130. Match the Column - I with Column - II and select the correct option from the codes given below.

Column - I		Column - II	
A.	Operator site	(i)	Binding site for RNA polymerase
B.	Promoter site	(ii)	Binding site for repressor molecule
C.	Regulator gene	(iii)	Codes for protein / enzyme
D.	Structural gene	(iv)	Codes for repressor molecule

- (1) A - (ii), B - (i), C - (iii), D - (iv)
 (2) A - (ii), B - (i), C - (iv), D - (iii)
 (3) A - (iv), B - (iii), C - (i), D - (ii)
 (4) A - (ii), B - (iii), C - (i), D - (iv)

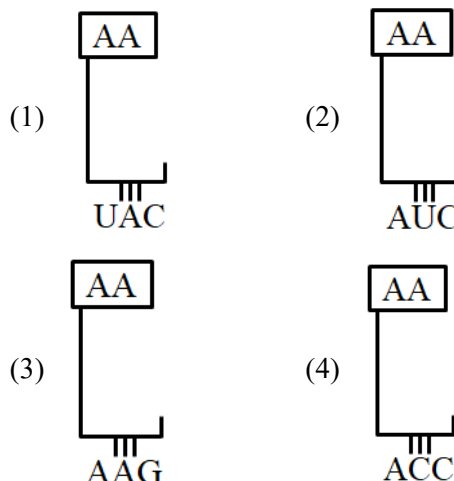
131. The process of transcription initiates when-

- (1) DNA dependent RNA polymerase reaches at the transcription start site.
 (2) DNA dependent RNA polymerase binds to promotor.
 (3) RNA dependent RNA polymerase binds to promotor.
 (4) RNA dependent DNA polymerase binds to promotor.

132. Which one of the following codon has dual functions.

- (1) GUA (2) ACU (3) AUG (4) UAG

133. Which of the following t-RNA not present in cell?



134. Genetic code is

- (1) Triplet, universal, ambiguous and degenerate
 (2) Triplet, universal, non-ambiguous and nondegenerate
 (3) Triplet, universal, non-ambiguous and degenerate
 (4) Triplet, universal, ambiguous and non-degenerate

135. Read the following four statements (A–D) :-

- (A) Newly arrived t–RNA with amino acid binds at A–site except at initiation codon.
 (B) Reverse transcription is catalysed by RNA dependent DNA polymerase.
 (C) Anti–shine delgarno sequence is present in larger subunit of ribosome.
 (D) The enzyme that catalyses the peptide bond formation is located in smaller ribosomal subunit.

How many of the above statement are correct ?

- (1) Four
 (2) Three
 (3) Two
 (4) One

SECTION-B (BIOLOGY-I)

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136. Which of the following is true about terminator?

- (1) The terminator is located towards 5' end of the coding strand
- (2) The terminator usually define the end of the process transcription
- (3) The terminator is located towards 3' end of the coding strand
- (4) Both (2) & (3)

137. 5' 7mG X AUG UAG Y Poly A tail 3'

In the given diagram for eukaryotic mRNA X and Y are respectively :

- (1) UTR and UTR
- (2) S.D. sequence and UTR
- (3) TATA box and UTR
- (4) TATA box and Terminator

138. In lac operon of *E.coli*, an operator gene combines with

- (1) inducer gene to switch on the transcription of structural gene.
- (2) regulator gene to switch on the transcription of structural gene.
- (3) repressor protein to switch off the transcription of structural gene.
- (4) repressor protein to switch on the transcription of structural gene.

139. Which set of RNA are involved in protein synthesis ?

- (1) tRNA, mRNA, rRNA
- (2) tRNA, mRNA, hnRNA
- (3) hnRNA, mRNA, rRNA
- (4) hnRNA, tRNA, rRNA

140. At the time of translation, r-RNA plays :-

- (1) Only structural role
- (2) Only catalytic role
- (3) Structural and catalytic role
- (4) Adapter role

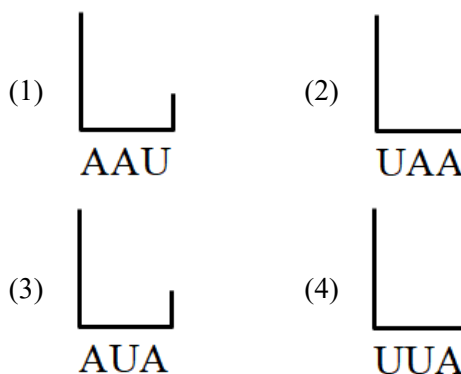
141. In lac operon, i-gene codes for

- (1) Inducer of lac operon
- (2) Repressor of lac operon
- (3) Hydrolysis enzyme
- (4) Permease

142. First triplet codon discovered was :

- (1) AUG
- (2) GUG
- (3) UUU
- (4) UAG

143. The tRNA for codon 'UUA' is :

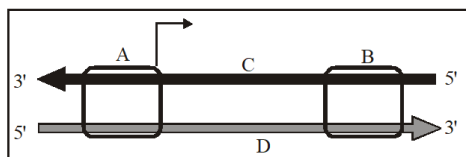


144. If the bacteria don't have lactose around them, they would no longer require the synthesis of enzyme ..A.. Therefore, in simple terms it is the metabolic, physiological or environmental condition that regulate the ...B....

Complete the statements filling the correct option in given blanks.

- (1) A-β-galactosidase; B-expression of genes
- (2) A-β-galactosidase; B-induction of lipid
- (3) A-β-galactosidase; B-repression of genes
- (4) A-maltase; B-repression of genes

145. Given below is the diagrammatic sketch of the transcription unit. Identify the parts labelled A, B, C and D. Select the right option about them :-



	Part-A	Part-B	Part-C	Part-D
(1)	Promoter	Pribnow box	Antisense strand	Sense strand
(2)	Terminator	Promoter	Sense strand	Antisense strand
(3)	Promoter	Terminator	Antisense strand	Sense strand
(4)	Promoter	Terminator	Sense strand	Antisense strand

146. Which enzyme unwinds the helix of DNA at the time of transcription ?

- (1) Helicase (2) Topoisomerase
(3) Gyrase (4) RNA polymerase

147. **Statement-I** : Single type of DNA dependent RNA polymerase can catalyse transcription of all types of RNA in bacteria.

Statement-II : RNA polymerase uses nucleotide triphosphates as substrate.

- (1) Both **Statement I** and **Statement II** are incorrect.
(2) **Statement I** is correct but **Statement II** is incorrect.
(3) **Statement I** is incorrect but **Statement II** is correct.
(4) Both **Statement I** and **Statement II** are correct.

148. **Assertion (A)** :- Genetic codes are degenerated.

Reason (R) :- Some amino acids are coded by more than one codon.

- (1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
(2) (A) is correct but (R) is not correct
(3) (A) is incorrect but (R) is correct
(4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

149. Match the column :-

Column-I		Column-II	
(A)	Termination	(i)	One codon codes for only one amino acid
(B)	Non-ambiguity	(ii)	One amino acid is coded by more than one codon
(C)	Universality	(iii)	Ochre, Amber, Opal
(D)	Degeneracy or redundancy	(iv)	From bacteria to human beings UUU codes for phenyl alanine

- (1) A = (iii), B = (ii), C = (iv), D = (i)
(2) A = (iii), B = (i), C = (iv), D = (ii)
(3) A = (iv), B = (i), C = (ii), D = (iii)
(4) A = (iii), B = (iv), C = (i), D = (ii)

150. Given below is a table of some disorders and type of mutation. Match disorder with correct mutation and choose the correct answer :-

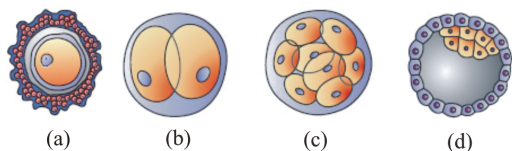
	Disorder		Mutation
(1)	Barr eye character	(a)	Reciprocal translocation
(2)	Cry du chat syndrome	(b)	Duplication
(3)	Thalassaemia	(c)	Substitution mutation
(4)	Sickle cell anaemia	(d)	Deletion
(5)	Chronic myeloid leukemia	(e)	Frame shift mutation

- (1) 1 - d, 2 - b, 3 - e, 4 - a, 5 - c
(2) 1 - a, 2 - b, 3 - c, 4 - d, 5 - e
(3) 1 - e, 2 - d, 3 - c, 4 - b, 5 - a
(4) 1 - b, 2 - d, 3 - e, 4 - c, 5 - a

Topic : MOLECULAR BASIS OF INHERITANCE (COMPLETE), HUMAN REPRODUCTION, REPRODUCTIVE HEALTH, HUMAN HEALTH AND DISEASE

SECTION-A (BIOLOGY-II)

151. The figure below show four stage (a, b, c, d) of human development. Select the option giving correct identification together with site of occurrence ?



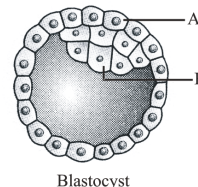
		Developmental stage	Site of occurrence
(1)	(a)	Fertilised egg	Isthmus part of fallopian tube
(2)	(b)	Zygote	End part of fallopian tube
(3)	(c)	Morula	Starting point of Fallopian tube
(4)	(d)	Blastocyst	Uterine cavity

152. Match the column-A and B about the embryonic development of human :-

	Column-A		Column-B
(i)	End of one month	(a)	Most of the major organ systems developed
(ii)	End of second month	(b)	Appearance of hair on head
(iii)	End of three month	(c)	Heart formation
(iv)	During fifth month	(d)	Eye lids separated
(v)	End of sixth month	(e)	Limbs and digits formatio

- (1) (i)-e, (ii)-b, (iii)-c, (iv)-a, (v)-d
 (2) (i)-e, (ii)-c, (iii)-b, (iv)-b, (v)-d
 (3) (i)-c, (ii)-e, (iii)-b, (iv)-a, (v)-d
 (4) (i)-c, (ii)-e, (iii)-a, (iv)-b, (v)-d

153. Identify A and B and their respective functions :



	A	B	Function of	
			A	B
(1)	Trophoblast	Inner cell mass	Attach to the Endometrium	Differentiated as embryo
(2)	Trophoblast	Inner cell mass	Differentiated as embryo	Attach to the Endometrium
(3)	Mesoderm	Inner cell mass	Differentiated as embryo	Attach to the Endometrium
(4)	Ectoderm	Mesoderm	Attach to the Endometrim	Differentiated as embryo

154. Identify the correct match from the columns I, II and III :-

	Column-I (Contraceptive method)	Column-II (Mechanism)	Column-III (Example)
(A)	Copper releasing IUDs	(a) Prevent ovulation	(i) LNG-20
(B)	Hormonal releasing IUDs	(b) Destroy sperms	(ii) Today
(C)	Daily oral pill	(c) Prevent implantation	(iii) MALA-D
(D)	Vaginal tablets	(d) Suppress sperm motility	(iv) Multiload 375

- (1) A – d – iv, B – a – i, C – c – iii, D – b – ii
 (2) A – b – iv, B – c – ii, C – a – i, D – c – iii
 (3) A – d – iv, B – c – i, C – a – iii, D – b – ii
 (4) A – d – iv, B – a – ii, C – c – iv, D – b – iii

155. Which one of the following is an in-vivo technique ?

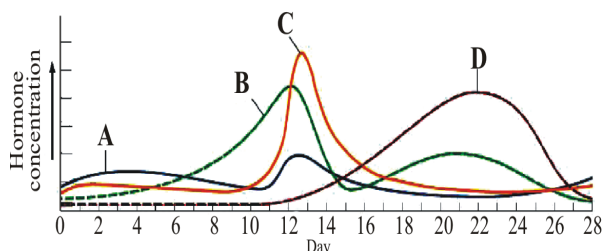
(1) ZIFT (2) IUT (3) ICSI (4) GIFT

156. Which of the following hormone is released by sertoli cells in response to excess spermatogenesis?

(1) GnRH (2) ICSH
 (3) FSH (4) Inhibin

157. How many of the following contraceptive method in the list given below have almost no side effects?
Diaphragm, Cervical cap, Pills, IUD, Lactational amenorrhea, Periodic abstinence, Depoprovera, Norplant, Coitus interruptus
- (1) Six (2) Three (3) Four (4) One
158. Arrange the following reproductive events occurring in human from first to last and find out the correct sequence :-
- (a) Insemination
(b) Parturition
(c) Formation of blastocyst
(d) Gametogenesis
(e) Gestation
(f) Fertilisation
- (1) dafceb (2) bcefad (3) becfad (4) dfabce
159. During menstrual cycle FSH surge is less than LH surge because of :
- (1) Estrogen (2) Progesterone
(3) Inhibin (4) Relaxin
160. During fertilization a membrane form in perivitelline space, which prevent polyspermy to egg. It is formed by :
- (1) Acrosome of sperm
(2) Centriole of ovum
(3) Cortical granules of oocyte
(4) Proximal centriole of sperm
161. In a female undergoing tubectomy, which of the following event will not occur ?
- (1) Completion of meiosis-II and formation of ovum
(2) Fertilization
(3) Menstruation
(4) Both 1 and 2
162. Which hormone regulates the GnRH pulse by feedback mechanism ?
- (1) Estrogen (2) Relaxin
(3) Inhibin (4) ABP
163. Primary peak level of estrogen and secondary peak level of estrogen in menstrual cycle occurs due to:-
- (1) Mature graafian follicle and corpus luteum respectively.
(2) Ovulation and corpus luteum
(3) Only formation of corpus luteum
(4) Ovulation and destruction of corpus luteum respectively.
164. How many statements are correct regarding interferons ?
- (A) These are secreted from bacterial infected cell.
(B) These are virus specific but not species specific.
(C) Interferons are an antiviral protein made up of 270 amino acid.
(D) Interferons can be used for prophylaxis and treatment of viral infections.
(E) α -interferon shows some success in regression of Kaposi Sarcoma.
- (1) 2
(2) 3
(3) 4
(4) 1
165. If estrogen is not able to attain peak level during pre-ovulatory phase of menstruation cycle in a female then what will be happen in this cycle?
- (1) Inhibit ovulation
(2) Corpus luteum will degenerate
(3) LH surge will not occur
(4) Both (1) and (3)
166. Which of the following event induced the completion of the meiotic division of the secondary oocyte ?
- (1) The process of contact of sperm with egg
(2) The entry of the sperm into the cytoplasm of the egg
(3) The release of ovum from the ovary
(4) Collection of ovum after ovulation by infundibulum

167. Identify the hormones A, B, C and D from the graph by observing their concentrations in blood during the menstrual cycle of a female :-



	A	B	C	D
(1)	FSH	Progesterone	Estrogen	LH
(2)	FSH	Estrogen	LH	Progesterone
(3)	FSH	LH	Progesterone	Estrogen
(4)	LH	Estrogen	Progesterone	FSH

168. Which of the following event will definitely confirm the pregnancy in female ?

- (1) Degeneration of stratum functionalis.
- (2) Presence of hCG in urine of female.
- (3) Delay of menstrual cycle
- (4) Both 1 & 2

169. How many of following structures may be present in fallopian tube of a female ?
Primary oocyte, Secondary spermatocyte, Zygote, First polar body, Morula, Secondary oocyte, Spermatozoa, Second polar body.

- (1) Six
- (2) Five
- (3) Four
- (4) Three

170. Which of the following is not correct regarding parturition ?

- (A) Decrease in estrogen progesterone ratio.
- (B) Complex neuro-endocrine mechanism
- (C) Signal originate from fully developed foetus only.
- (D) Oxytocin is main parturition hormone.
- (E) Degeneration of ovary after parturition

- (1) only E
- (2) A and E
- (3) A and C
- (4) A, C and E

171. Which hormone increases manifold in maternal blood due to significant changes in metabolism ?

- (1) Estrogen
- (2) LH
- (3) Relaxin
- (4) Thyroxine

172. **Assertion** : Lactational amenorrhea is an example of natural method of contraception.

Reason : High level of prolactin suppresses the secretion of gonadotropins.

- (1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**
- (2) **Assertion** is true but **Reason** is false
- (3) **Assertion** is false but **Reason** is true
- (4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**

173. **Statement-I** : Gonadotropin hormones act on testis and ovary and are steroid in nature.

Statement-II : At puberty spermatogenesis and oogenesis start in the presence of gonadotropin.

- (1) Statement-I is correct but statement-II is false.
- (2) Statement-I is incorrect but statement-II is true.
- (3) Both Statement-I and statement-II are true.
- (4) Both Statement-I and statement-II are false.

174. How many primary spermatocytes and primary oocytes are required to form 400 sperms & 400 ova respectively ?

- (1) 100 and 400
- (2) 100 and 200
- (3) 100 and 100
- (4) 400 and 400

175. **Assertion** : Follicular development do not occur in the luteal phase of the menstrual cycle.

Reason : Gonadotropins levels are low in the luteal phase.

- (1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**
- (2) **Assertion** is true but **Reason** is false
- (3) **Assertion** is false but **Reason** is true
- (4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**

176. Which of the following is correctly matched pair regarding innate immunity ?

(A)	Physiological barrier	Acid in stomach, tears from eye
(B)	Physical barrier	Skin and mucus coating of the gastro-intestinal tract
(C)	Cellular barrier	B and T-lymphocytes
(D)	Cytokine barrier	PMNL Neutrophils & certain leucocytes
(E)	Cellular barrier	Monocytes and natural killer cells

- (1) A, B and C
- (2) A, B, D and E
- (3) A, B and E
- (4) A, B, C and E

177. Which of the following sexually transmitted disease is incorrectly matched with its causative agent ?

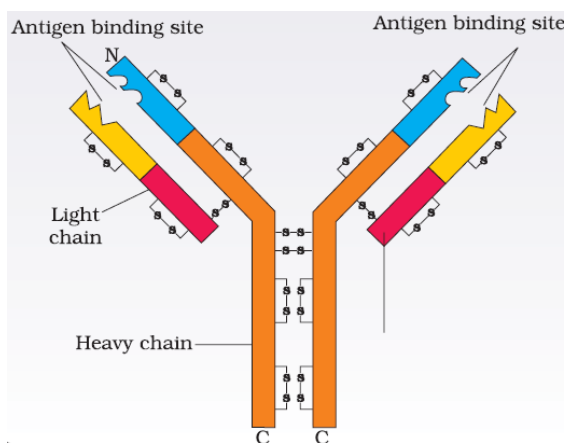
- (A) Syphilis - *Treponema pallidum*
 (B) Genital herpes - *Trichomonas vaginalis*
 (C) Genital warts - Herpes simplex virus
 (D) Gonorrhoea - *Neisseria gonorrhoeae*
 (E) Chlamydia - *Chlamydia trachomatis*

- (1) A, B and C (2) B and C
 (3) A and C (4) only C

178. IgA received from colostrum is an example of :

- (1) Natural passive immunity
 (2) Innate immunity
 (3) Natural active immunity
 (4) Artificial active immunity

179. Choose the correct statements regarding the structure given in the below diagram.



- (A) The given structure is of a glycoprotein molecule.
 (B) The molecule can be represented as H_2L_2 molecule
 (C) The given molecule comprises major part of our innate immunity.
 (D) The molecule is synthesized by T-lymphocytes in our body.
 (E) The response generated by this molecule is called as humoral immune response.

- (1) A, B, C, D and E (2) A, B, C and E
 (3) A, B, D and E (4) A, B and E

180. Which of the following venereal disease is/are viral disease and donot affect specifically reproductive organs ?

- (A) AIDS
 (B) Genital herpes
 (C) Hepatitis-B
 (D) Syphilis
 (E) Genital warts

- (1) A and B (2) A, B, C and E
 (3) A and C (4) Only A

181. Which of the following statements are not correct regarding human immune system ?

- (A) Bone marrow and thymus are the primary lymphoid organs.
 (B) Innate immune system is very specific and uses different macrophages.
 (C) Cell mediated immune system is responsible for rejection of kidney graft.
 (D) Antibodies are glycoprotein molecules, each of which has four heavy chains.
 (E) MALT constitutes about 50 percent of the lymphoid tissue in human body.

- (1) A, C, D and E (2) B and D
 (3) B, C and D (4) Only B

182. Which of the following statements are not correct regarding immune system of higher vertebrates ?

- (A) The human immune system consists of lymphoid organs, tissue, cells and insoluble fat molecules like antibodies.
 (B) Memory based acquired immunity is characteristic of higher vertebrates.
 (C) The antibodies are found in blood and they generate humoral immune response.
 (D) The primary lymphoid organs are organs which provide the sites for interaction of lymphocytes with the antigen.
 (E) Lymph nodes serve to trap the micro-organism or antigen which are present in tissue fluid.

- (1) D and E (2) Only D
 (3) A and D (4) A, D and E

183. Human immune system consists of lymphoid organs, tissues, cells and molecules like antibodies. Choose the wrong pair from the given below immune system constituents.

(A)	Thymus	Primary lymphoid organ, lobed organ, present near heart.
(B)	Peyer patches	Secondary lymphoid organ, present in small intestine
(C)	Spleen	Primary lymphoid organ, large bean shaped
(D)	Appendix	Secondary lymphoid organ, provide site for interaction of lymphocytes with antigen.
(E)	Lymph node	Solid structure at different point of lymphatic system, serves to trap the antigen.

- (1) D and E
(2) C and D
(3) B and E
(4) Only C
184. AIDS is a venereal disease which can transmit through :
- (a) Using sterile needles
(b) Transfusion of blood from infected person.
(c) Infected mother to foetus
(d) Kissing
(e) Inheritance
- Choose the correct answer from the options given below :
- (1) (a), (b) and (c) only
(2) (b), (c) and (d) only
(3) (b) and (c) only
(4) (a) and (c) only

185. If in a couple fertilisation is failed and AI technique is already done then which advance technique will be preferred next ?
- (1) ET (2) GIFT (3) ICSI (4) ZIFT

SECTION-B (BIOLOGY-II)

186. Fertilisation has following processes :-
- (a) Plasmogamy
(b) Karyogamy
(c) Syngamy
(d) Amphimixis
- Arrange these in correct sequence :
- (1) $a \rightarrow b \rightarrow c \rightarrow d$ (2) $b \rightarrow a \rightarrow d \rightarrow c$
(3) $c \rightarrow a \rightarrow b \rightarrow d$ (4) $c \rightarrow a \rightarrow d \rightarrow b$
187. Which of the following contraceptive method has its additional benefit of protecting the user from contracting STD's and AIDS ?
- (1) IUDs (2) Condom
(3) Hormonal Oral pill (4) Implant
188. Which of the following method of contraception is effective only upto a maximum period of six months following parturition ?
- (1) LNG-20
(2) Coitus interrupts
(3) Lactational amenorrhoea
(4) CuT
189. How many of the following cells related to spermatogenesis are diploid in nature ?
- Spermatogonia, Primary Spermatocyte, Secondary Spermatocyte, Spermatid, Sperm.
- (1) Four (2) Three (3) Two (4) One
190. How many seminiferous tubules are present in each testicular lobule ?
- (1) 250 (2) 750 -1000
(3) 2 - 3 (4) 100

191. Read the given statements carefully :-

- (a) 'A' is single.
 - (b) The shape of 'A' is like an inverted pear.
 - (c) 'A' opens into vagina
 - (d) The wall of 'A' has three layers of tissue.
- Here 'A' is

- (1) Ovary (2) Uterus
- (3) Urethra (4) Fallopian tube

192. Colostrum, the first milk secretion of mammary gland is rich in Immunoglobulin :

- (1) IgE (2) IgM (3) IgA (4) IgG

193. Which of the following is not the property of primary immune response ?

- (1) Anamnestic
- (2) Slow
- (3) Weak
- (4) Less antibodies formed

194. **Assertion** : Endometrium is necessary for implantation of the fertilised ovum.

Reason : A large amount of progesterone is secreted by corpus Luteum.

- (1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.
- (2) **Assertion** is true but **Reason** is false
- (3) **Assertion** is false but **Reason** is true.
- (4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

195. Identify the hormone which act on the uterine muscle and causes stronger uterine contractions.

- (1) Estrogen
- (2) Progesterone
- (3) Oxytocin
- (4) Relaxin

196. **Assertion** : Sterilisation procedure in the male is called tubectomy.

Reason : In vasectomy, a small part of the vas deferens is removed or tied up to block gamete transport.

- (1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**
- (2) **Assertion** is true but **Reason** is false
- (3) **Assertion** is false but **Reason** is true
- (4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**

197. Match column-I and II with the correct options given below.

	Column-I		Column-II
(a)	MALT	(i)	Secondary lymphoid organ
(b)	Bone marrow	(ii)	Cells of immune system
(c)	Monocytes	(iii)	Lymphoid tissue
(d)	Glyco-protein molecules	(iv)	Primary lymphoid organ
(e)	Peyer's patches of small intestine	(v)	Immunity providing molecules.

- (1) a-iv, b-iii, c-i, d-v, e-ii
- (2) a-iii, b-iv, c-v, d-ii, e-i
- (3) a-iv, b-i, c-ii, d-v, e-iii
- (4) a-iii, b-iv, c-ii, d-v, e-i

198. ___A___ is the main lymphoid organ where all blood cells including lymphocytes are produced, while ___B___ is a lobed organ located near the heart and beneath the breast bone. Fill the blanks with the most suitable options :

- (1) A-Thymus, B-Bone marrow
- (2) A-Bone marrow, B-Thymus
- (3) A-Bone marrow, B-Spleen
- (4) A-Spleen, B-Thymus

199. Which of the following statements given below regarding sexually transmitted disease are incorrect ?
- (A) Sexually transmitted disease can spread through kissing and inheritance.
- (B) AIDS and Hepatitis-B both are sexually transmitted diseases that do not specifically affect reproductive organs.
- (C) Genital herpes is a non-curable STD.
- (D) STD can lead to infertility, abortions and even cancer of reproductive tracts.

- (1) only A (2) A and B
- (3) A, B and C (4) Only B

200. Choose the incorrect statement regarding antibody molecule ?

- (1) Antibody are glycoprotein molecule containing globulin protein
- (2) Antibody is represented as H_2L_2 molecule.
- (3) Antibody has two polypeptide chains.
- (4) Antibodies are secreted by B-cells.

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